Finite-Rod Model and Validity Checks: Fin Equation, Finite-Length Error, and Transverse Biot Number

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This assignment builds the stationary finite-rod (fin) model used for the Module 7 course rod and makes two provisional validity predictions before heating. Applied to Lienhard Problem 1.3, Fourier's law gives temperature drops from 0.1 K to 2000 K across a 1-cm slab at the same heat flux; the three low-conductivity cases are physically unreasonable. An energy balance on a rod slice gives the fin equation $d^2\theta/dx^2 = m^2\theta$ with $m^2 = hP/(kA)$, whose insulated-tip solution is $\cosh[m(L-x)]/\cosh mL$. For the 6061-T6 rod ($d = 12.70$ mm, $L = 26.90$ cm, $k = 166$–172 $\mathrm{W\,m^{-1}K^{-1}}$) and the planning value $h = 10$ $\mathrm{W\,m^{-2}K^{-1}}$, $m = 4.34$ $\mathrm{m^{-1}}$ and $mL = 1.17$, so the semi-infinite profile e^{-mx} is in error by 11–30% at the three farthest planned sensors: the rod is not expected to be semi-infinite over the measured region. In contrast, $\mathrm{Bi}_\perp = hd/(4k) = 1.9 \times 10^{-4} \ll 0.1$, so each cross section can be represented by one temperature.

Assignment A4 (individual). Repository: `https://github.com/tyin-source/__Lab_Module_7__`; Git commit `94ede0416ea097463bf811f10fbb121f1848e46b`.

Parts 3 and 4 are provisional predictions recorded before heating the rod. They use the measured rod geometry, the planned sensor positions, a literature range for k, and the planning value $h = 10$ $\mathrm{W\,m^{-2}K^{-1}}$ given in the Part III handout [2]. Equation numbers in parentheses such as (4.28) are those of Lienhard [1].

PART 1. APPLY FOURIER'S LAW: LIENHARD PROBLEM 1.3

Problem statement

A flat slab of thickness $L = 0.01$ m carries a steady heat flux $q = 5000$ $\mathrm{W/m^2}$ from its hot side to its cold side. The cold side is held at $T_c = -40\,^\circ\mathrm{C}$. For each material in Table I, compute the temperature drop ΔT across the slab, find the hot-side temperature $T_h = T_c + \Delta T$, and state whether the case is physically reasonable and why.

Derivation from Fourier's law

Start from the one-dimensional Fourier law,

$$q_x = -k\frac{dT}{dx}.$$

Assumptions:

1. **Steady state**: no time dependence, $\partial T/\partial t = 0$;
2. **One-dimensional conduction**: $T = T(x)$ only;
3. **Constant** k: thermal conductivity independent of temperature over the range;
4. **No internal heat generation**.

With no source and steady state, the energy balance on a slice gives

$$\frac{dq_x}{dx} = 0 \quad \Longrightarrow \quad q_x = q = \text{constant}.$$

Then

$$\frac{dT}{dx} = -\frac{q}{k} = \text{constant},$$

so the temperature profile is **linear in** x. This linearity is a *result* of the steady, source-free, constant-k assumptions, not an extra assumption. Integrating from $x = 0$ (hot side) to $x = L$ (cold side),

$$T(L) - T(0) = -\frac{qL}{k}.$$

Defining $\Delta T = T_h - T_c = T(0) - T(L)$,

$$\boxed{\Delta T = \frac{qL}{k}} \qquad \boxed{T_h = T_c + \frac{qL}{k}}.$$

Unit check

$$\frac{\mathrm{W/m^2 \cdot m}}{\mathrm{W/(m \cdot K)}} = \frac{\mathrm{W/m}}{\mathrm{W/(m \cdot K)}} = \mathrm{K}.$$

A temperature difference in kelvin equals the same difference in °C, so T_h can be reported in °C.

Worked example: silver

For silver, $k = 427$ $\mathrm{W\,m^{-1}K^{-1}}$. With $q = 5000$ $\mathrm{W/m^2}$ and $L = 0.01$ m,

$$\Delta T = \frac{(5000\ \mathrm{W/m^2})(0.01\ \mathrm{m})}{427\ \mathrm{W/(m \cdot K)}} = \frac{50}{427}\ \mathrm{K}$$
$$\approx 0.117\ \mathrm{K}.$$

Hence

$$T_h = -40\,^\circ\mathrm{C} + 0.117\ \mathrm{K} \approx -39.88\,^\circ\mathrm{C}.$$

The hot side stays far below silver's melting point, and the temperature range is tiny, so constant-k, steady, one-dimensional conduction is an excellent approximation.

Table calculation for all materials

For every material the same formula applies,

$$\Delta T = \frac{qL}{k} = \frac{50}{k}\ \mathrm{K}, \qquad T_h = -40\,^\circ\mathrm{C} + \Delta T,$$

with the results in Table I. The metals use the 20 °C column, as the book's silver answer does. With k interpolated to −40 °C, silver gives 0.1165 K and aluminum 0.2097 K, a change below 1%.

Which cases are unreasonable, and why

(a) Compare T_h with each material's physical or service limit.

- **Ice:** $T_h \approx -17.4\,^\circ\mathrm{C}$, still below the 0 °C melting point. Ice remains solid, which is consistent.
- **Spruce:** $T_h \approx 415\,^\circ\mathrm{C}$. Wood chars and ignites at roughly 250–300 °C, so spruce cannot exist at this temperature. **Unreasonable.**
- **85% magnesia:** $T_h \approx 706\,^\circ\mathrm{C}$. The $k = 0.067\ \mathrm{W\,m^{-1}K^{-1}}$ entry is **not** "MgO"; Lienhard Table A.2 lists "85% magnesia" separately from "Magnesium oxide, polycrystalline" ($k = 48\ \mathrm{W\,m^{-1}K^{-1}}$). 85% magnesia is a magnesium-carbonate-based pipe insulation with a documented service limit of **up to 315** °C [3]. The computed hot side is more than twice that limit, so this case is **unreasonable**.
- **Aerogel:** $T_h \approx 2043\,^\circ\mathrm{C}$. Silica aerogel is a porous SiO_2 network. The melting point of SiO_2 is about **1710** °C [4], with reported values of roughly **1600**–**1725** °C [5]. 2043 °C is above the melting point of the solid skeleton itself, which is a stronger objection than sintering. **Unreasonable.**

(b) Is constant k still credible when ΔT is hundreds or thousands of kelvin? For the three metals and ice, ΔT is at most a few tens of kelvin, so constant k is fine. For spruce, 85% magnesia, and aerogel, ΔT is several hundred to over two thousand kelvin. Over such ranges k is not constant: it varies with temperature, and for porous insulators radiation across pores adds a strongly temperature-dependent contribution. The linear profile derived above no longer holds, so the computed T_h is only a nominal estimate.

The 85% magnesia entry makes this concrete: Lienhard Table A.2 gives $k = 0.067$ at 38 °C and $k = 0.08$ at 204 °C. Even within the material's rated range, k rises by about 20%. Extrapolating a single constant $k = 0.067$ to 706 °C is therefore not defensible.

(c) Connection to TEC over-temperature protection. A thermoelectric cooler uses the Peltier effect to **pump heat from the cold side to the hot side**. The element's low k is actually helpful for the cooling function: it reduces heat conduction back from the hot side to the cold side. The over-temperature risk is **not** that the element's own k is low; it lies on the **heat-rejection path** on the hot side.

The hot side must reject both the pumped heat and the electrical power through the heat sink. If the thermal resistance of that path becomes large (a failed fan, poor thermal contact, a blocked heat sink), the hot-side temperature rise is

$$\Delta T_{\text{hot side}} \approx \dot{Q}_{\text{reject}}\, R_{\text{path}}.$$

This has the **same form** as the slab result $\Delta T = qL/k$: a fixed heat flow through a large thermal resistance produces a large temperature rise. This is exactly why Module 4 includes a series thermal switch that opens around 70 °C: it interrupts the path before the hot side overheats. The slab calculation is a minimal illustration of the same warning: a large ΔT can appear when a fixed heat flux is forced through a high-thermal-resistance path.

PART 2. BUILD THE STATIONARY FINITE-ROD MODEL

2.1 Stationary fin equation

Assumptions.

1. Steady state: $\partial T/\partial t = 0$, so no slice stores energy.
2. No internal heat generation in the rod.
3. k, A, and P are constant along the rod (a uniform cylinder).
4. h is uniform over the side surface and T_∞ is constant. Radiation is neglected or linearized into h.
5. Each cross section has a single temperature $T(x)$. This requires a small transverse Biot number, $\mathrm{Bi}_\perp = h(A/P)/k \ll 1$. For a circular rod, $A/P = (\pi d^2/4)/(\pi d) = d/4 = r/2$, so $\mathrm{Bi}_\perp = hr/(2k) \ll 1$.

TABLE I. Lienhard Problem 1.3: $\Delta T = qL/k$ and T_h for $q = 5000$ W/m^2, $L = 0.01$ m, $T_c = -40\,^\circ$C. Conductivities from Lienhard Tables A.1 and A.2 [1]; values from `python/analysis/a4_problem_1_3.py`.

Material	k (W/(m K))	Source	ΔT (K)	T_h (°C)	Reasonable?
Pure silver (99.99+%)	427	Table A.1, 20 °C	0.117	−39.88	Yes: tiny ΔT, far from any limit
Pure aluminum	237	Table A.1, 20 °C	0.211	−39.79	Yes: tiny ΔT, far from any limit
Mild steel (0.5% carbon)	54	Table A.1, 20 °C	0.926	−39.07	Yes: tiny ΔT, far from any limit
Ice	2.215	Table A.2, 0 °C	22.6	−17.4	Yes: $T_h < 0\,^\circ$C, ice stays solid
Spruce	0.11	Table A.2, 20 °C	455	415	**No**: wood chars and ignites well below this
Insulation (85% magnesia, *not* MgO)	0.067	Table A.2, 38 °C	746	706	**No**: far above its service limit
Silica aerogel	0.024	Table A.2, 0 °C	2083	2043	**No**: above the SiO_2 melting point

Energy balance. Take a slice $[x,\ x+\delta x]$:

- heat conducted in through the left face (**Fourier's law**, $q = -k\,dT/dx$, over the area A): $-kA\,\dfrac{dT}{dx}\Big|_x$;
- heat conducted out through the right face (**Fourier's law**): $-kA\,\dfrac{dT}{dx}\Big|_{x+\delta x}$;
- heat lost from the side (**Newton's law of cooling**, $q = h(T - T_\infty)$, over the side area $P\,\delta x$): $hP\,\delta x\,(T - T_\infty)$.

Conservation of energy for the slice at steady state (no storage, no generation): setting in − out − side loss = 0 gives

$$\left(-kA\frac{dT}{dx}\right)_x - \left(-kA\frac{dT}{dx}\right)_{x+\delta x} - hP\,\delta x\,(T - T_\infty) = 0. \quad (4.28)$$

Dividing by δx and letting $\delta x \to 0$,

$$kA\,\frac{d^2T}{dx^2} = hP\,(T - T_\infty).$$

Because T_∞ is constant, $d^2T/dx^2 = d^2(T - T_\infty)/dx^2$. Therefore

$$\frac{d^2(T - T_\infty)}{dx^2} = m^2\,(T - T_\infty), \qquad m^2 = \frac{hP}{kA}. \quad (4.30)$$

For a circular rod, $m^2 = 4h/(kd)$.

Unit check. Every term in (4.28) has units of W:

- conduction term: $kA\,dT/dx$ has units $[\mathrm{W\,m^{-1}K^{-1}}][\mathrm{m^2}][\mathrm{K\,m^{-1}}] = \mathrm{W}$;
- side-loss term: $hP\,\delta x\,(T - T_\infty)$ has units $[\mathrm{W\,m^{-2}K^{-1}}][\mathrm{m}][\mathrm{m}][\mathrm{K}] = \mathrm{W}$.

Both sides of $kA\,d^2T/dx^2 = hP\,(T - T_\infty)$ are heat flow per unit length, W m^{-1}:

- left: $[\mathrm{W\,m^{-1}K^{-1}}][\mathrm{m^2}][\mathrm{K\,m^{-2}}] = \mathrm{W\,m^{-1}}$;
- right: $[\mathrm{W\,m^{-2}K^{-1}}][\mathrm{m}][\mathrm{K}] = \mathrm{W\,m^{-1}}$.

Both sides of (4.30) have units of K m^{-2}: the left side is K m^{-2}, and on the right m^2 has units

$$\frac{[\mathrm{W\,m^{-2}K^{-1}}][\mathrm{m}]}{[\mathrm{W\,m^{-1}K^{-1}}][\mathrm{m^2}]} = \mathrm{m^{-2}},$$

which multiplied by K gives K m^{-2}. $1/m$ has units of m and is the decay length. In an infinite rod, $T - T_\infty$ falls to $1/e$ of its value over each distance $1/m$.

2.2 Boundary conditions and their physical meaning

A second-order equation needs two boundary conditions. The course rod uses a Dirichlet condition at the base and a Neumann condition at the tip.

Base (Dirichlet): $T(0) = T_0$. Physically, this fixes the temperature at $x = 0$ to a known value. On the course rod, $x = 0$ is the first thermistor hole, and T_0 is the temperature that thermistor measures. It is not the TEC setpoint, because the contact interface and a length of rod lie between the TEC and the first hole, so the two temperatures differ. Nor is $x = 0$ the physical end of the rod. Choosing this interior plane as the analysis base is valid because the equation holds for any section of rod from that plane to the tip.

Tip (Neumann, adiabatic): $dT/dx|_{x=L} = 0$ (4.31b). Physically, no heat crosses the end face: $\dot{Q}(L) = -kA\,dT/dx|_L = 0$. This is an approximation. The real end face also loses heat by convection, which is described by the Robin condition (4.31a),

$$-kA\,\frac{dT}{dx}\Big|_{x=L} = h_L A\,[T(L) - T_\infty].$$

The approximation is reasonable because the ratio of end-face area to side area is

$$\frac{\pi d^2/4}{\pi dL} = \frac{d}{4L} \ll 1,$$

so the end face carries only a small fraction of the total heat loss. Part 3 quantifies the difference between the two solutions (Example 4.8 and Problem 4.20).

2.3 Finite insulated-tip solution

Nondimensionalization. Let $\Theta = (T-T_\infty)/(T_0-T_\infty)$ and $\xi = x/L$. Then (4.30) becomes

$$\frac{d^2\Theta}{d\xi^2} = (mL)^2\,\Theta. \tag{4.34}$$

The boundary conditions become $\Theta(0) = 1$ and $d\Theta/d\xi|_{\xi=1} = 0$. The general solution and its derivative are

$$\Theta = C_1 e^{mL\xi} + C_2 e^{-mL\xi},$$
$$\frac{d\Theta}{d\xi} = mL\left(C_1 e^{mL\xi} - C_2 e^{-mL\xi}\right).$$

Solving for the constants. The base condition gives

$$C_1 + C_2 = 1. \tag{a}$$

The tip condition gives

$$mL\left(C_1 e^{mL} - C_2 e^{-mL}\right) = 0 \;\Rightarrow\; C_2 = C_1 e^{2mL}. \tag{b}$$

Substituting (b) into (a),

$$C_1\left(1 + e^{2mL}\right) = 1 \;\Rightarrow\; C_1 = \frac{1}{1+e^{2mL}} = \frac{e^{-mL}}{e^{-mL}+e^{mL}} = \frac{e^{-mL}}{2\cosh mL}.$$

Returning to (b),

$$C_2 = C_1 e^{2mL} = \frac{e^{mL}}{2\cosh mL}.$$

Combining into a hyperbolic cosine.

$$\Theta = \frac{e^{-mL}e^{mL\xi} + e^{mL}e^{-mL\xi}}{2\cosh mL} = \frac{e^{-mL(1-\xi)} + e^{mL(1-\xi)}}{2\cosh mL} = \frac{\cosh\left[mL(1-\xi)\right]}{\cosh mL}. \tag{4.41}$$

In dimensional form,

$$T(x) - T_\infty = (T_0 - T_\infty)\,\frac{\cosh\left[m(L-x)\right]}{\cosh mL}.$$

Verification.

- Differential equation: $\dfrac{d^2\Theta}{d\xi^2} = (mL)^2\dfrac{\cosh[mL(1-\xi)]}{\cosh mL} = (mL)^2\Theta$ ✓
- Base condition: $\Theta(0) = \cosh mL/\cosh mL = 1$ ✓
- Tip condition: $\dfrac{d\Theta}{d\xi} = -mL\,\dfrac{\sinh[mL(1-\xi)]}{\cosh mL}$, which at $\xi = 1$ equals $-mL\,\sinh 0/\cosh mL = 0$ ✓

Base heat rate and tip temperature. The base heat rate is

$$\dot{Q}_0 = -kA\frac{dT}{dx}\Big|_0 = -\frac{kA(T_0-T_\infty)}{L}\frac{d\Theta}{d\xi}\Big|_0 = kAm\,(T_0-T_\infty)\tanh mL = \sqrt{hPkA}\,(T_0-T_\infty)\tanh mL.$$

Unit check: $\sqrt{hPkA}$ has units

$$\sqrt{[\mathrm{W\,m^{-2}K^{-1}}][\mathrm{m}][\mathrm{W\,m^{-1}K^{-1}}][\mathrm{m^2}]} = \mathrm{W\,K^{-1}},$$

and multiplying by K gives W ✓. The tip temperature is

$$\Theta(1) = \frac{1}{\cosh mL}.$$

PART 3. PROVISIONAL FINITE-LENGTH PREDICTION

3.1 Problem 4.12: the semi-infinite limit

Problem 4.12 asks for Eq. (4.50) from the general solution (4.35), using the boundary conditions $T(x=0) = T_0$ and $T(x=L) = T_\infty$. The far end is held at room temperature; it is not insulated. With $\theta = T - T_\infty$, these are $\theta(0) = \theta_0$ and $\theta(L) = 0$, and the fin equation $\theta'' = m^2\theta$ has the general solution

$$\theta = C_1 e^{mx} + C_2 e^{-mx}.$$

The two conditions give $C_1 + C_2 = \theta_0$ and $C_1 e^{mL} + C_2 e^{-mL} = 0$. The second gives $C_2 = -C_1 e^{2mL}$; substituting into the first,

$$C_1 = -\frac{\theta_0\, e^{-mL}}{2\sinh mL}, \qquad C_2 = \frac{\theta_0\, e^{mL}}{2\sinh mL},$$

so

$$\frac{\theta(x)}{\theta_0} = \frac{e^{m(L-x)} - e^{-m(L-x)}}{2\sinh mL} = \frac{\sinh\left[m(L-x)\right]}{\sinh mL}.$$

This satisfies $\theta(0)/\theta_0 = 1$ and $\theta(L) = 0$. For fixed x, write it as

$$\frac{\theta(x)}{\theta_0} = e^{-mx}\,\frac{1 - e^{-2m(L-x)}}{1 - e^{-2mL}} \xrightarrow{L\to\infty} e^{-mx}, \tag{4.50}$$

the semi-infinite profile.

Physical meaning. The far-end condition enters only through the factor $e^{-2m(L-x)}$, which decays exponentially with the distance from the far end measured in units of $1/m$. A few decay lengths away from the far end, the temperature no longer depends on what happens there: the insulated tip (Part 2, $\cosh[m(L-x)]/\cosh mL$) and the room-temperature tip ($\sinh[m(L-x)]/\sinh mL$) both approach e^{-mx}. The far-end condition is effectively forgotten, which is why (4.50) is a limit of the finite solution rather than a new physical law.

3.2 Example 4.8 and Problem 4.20: the tip-boundary approximation

For a convecting tip with coefficient h_L, the exact solution (Lienhard Eqs. 4.48 and 4.49) is

$$\Theta = \frac{\cosh\left[mL(1-\xi)\right] + \dfrac{\mathrm{Bi}_{\mathrm{tip}}}{mL}\sinh\left[mL(1-\xi)\right]}{\cosh mL + \dfrac{\mathrm{Bi}_{\mathrm{tip}}}{mL}\sinh mL},$$

$$\frac{\dot{Q}_0}{\sqrt{hPkA}\,(T_0 - T_\infty)} = \frac{\tanh mL + h_L/(mk)}{1 + \big(h_L/(mk)\big)\tanh mL},$$

with $\mathrm{Bi}_{\mathrm{tip}} = h_L L/k$ (Lienhard's $\mathrm{Bi}_{\mathrm{ax}}$) and $\mathrm{Bi}_{\mathrm{tip}}/(mL) = h_L/(mk)$. Setting $h_L \to 0$ recovers the insulated-tip results (4.41) and (4.44).

Example 4.8. An aluminum rod with $d = 2.0$ cm, $k = 205\ \mathrm{W\,m^{-1}K^{-1}}$, and $L = 8.0$ cm protrudes from a 150 °C wall into 26 °C air with $h = h_L = 120\ \mathrm{W\,m^{-2}K^{-1}}$. Then $mL = 0.8656$, $\mathrm{Bi}_{\mathrm{ax}} = 0.0468$, and $h/(mk) = 0.0468/0.8656 = 0.0541$. The exact tip temperature is $\Theta_{\mathrm{tip}} = 0.6886$ ($T_{\mathrm{tip}} = 111.4\,^\circ\mathrm{C}$), while the insulated tip gives $\Theta_{\mathrm{tip}} = 1/\cosh(0.8656) = 0.7150$ ($T_{\mathrm{tip}} = 114.7\,^\circ\mathrm{C}$). The insulated tip overpredicts the tip excess temperature by about 3.8%.

Problem 4.20. With $M = \sqrt{hPkA}\,(T_0 - T_\infty)$, an insulated tip gives $\dot{Q}_{\mathrm{ins}} = M\tanh(mL)$, and a convecting tip gives

$$\dot{Q}_{\mathrm{conv}} = M\,\frac{\sinh(mL) + \frac{h}{mk}\cosh(mL)}{\cosh(mL) + \frac{h}{mk}\sinh(mL)}.$$

Numerically,

$$\frac{\dot{Q}_{\mathrm{ins}}}{M} = \tanh(0.8656) = 0.6992,$$
$$\frac{\dot{Q}_{\mathrm{conv}}}{M} = \frac{\sinh(0.8656) + 0.0541\cosh(0.8656)}{\cosh(0.8656) + 0.0541\sinh(0.8656)} = 0.7258,$$

so the insulated-tip approximation underpredicts the exact heat flow by

$$\frac{\dot{Q}_{\mathrm{conv}} - \dot{Q}_{\mathrm{ins}}}{\dot{Q}_{\mathrm{conv}}} = \frac{0.7258 - 0.6992}{0.7258} = 0.0367,$$

or **3.67%**, with the approximate heat flow slightly lower than the exact value.

Course rod. The same comparison with $h_L = h = 10\ \mathrm{W\,m^{-2}K^{-1}}$, $k = 167\ \mathrm{W\,m^{-1}K^{-1}}$, and the inputs of 3.3 gives $mL = 1.168$, $\mathrm{Bi}_{\mathrm{tip}} = hL/k = 0.0161$, and $h/(mk) = 0.0138$, four times smaller than in Example 4.8 (Table II). The insulated-tip boundary condition therefore changes the predicted course-rod profile by about 1% at most, at the tip itself.

3.3 Inputs

The inputs are listed in Table III.

Planned sensor positions. T1–T5 go in the holes at $x = 0$, 5.0, 10.2, 15.3, and 20.4 cm. The reasons, using the predictions of 3.4 for $L = 26.90$ cm and $h = 10\ \mathrm{W\,m^{-2}K^{-1}}$:

1. T1 must be at $x = 0$: the first hole is the analysis base defined in the handout.
2. The farthest hole, 20.4 cm, is essential. The decay length $1/m \approx 23$ cm is longer than the measured region, and even there the predicted Θ is only about 0.59, so the whole available range is needed to measure a large enough temperature decay.
3. The points are spread evenly on a plot of $\ln\theta$ against x, as the handout asks, rather than clustered at one end: the spacings are all about 5 cm, and the predicted $\ln\Theta$ values are 0, −0.17, −0.33, −0.45, and −0.53.
4. The set tests the difference between the finite and semi-infinite models: 5.0 cm is the last hole that meets the 5% criterion ($\epsilon_T = 4.6\%$), while at 10.2, 15.3, and 20.4 cm ϵ_T is 11%, 20%, and 30%, so the two fits should differ visibly in their residuals.
5. The holes at 1.2 and 2.4 cm are not used: their 1.2 cm spacing is only about ten times the ±1.3 mm position uncertainty, and neighboring points would differ little in temperature, adding little information.
6. The farthest sensor still has a large signal: with the base about 25 K above room temperature, the predicted excess temperature at 20.4 cm is about 15 K, far above the noise.

3.4 m, mL, and $\epsilon_T(x)$ at each planned sensor

For the circular rod, $P/A = 4/d$, so

$$m = \sqrt{\frac{4h}{kd}} = \sqrt{\frac{4(10\ \mathrm{W\,m^{-2}K^{-1}})}{(167\ \mathrm{W\,m^{-1}K^{-1}})(0.01270\ \mathrm{m})}} = 4.34\ \mathrm{m^{-1}}.$$

TABLE II. Insulated versus convecting tip for the course rod ($h_L = h = 10\ \mathrm{W\,m^{-2}K^{-1}}$, $k = 167\ \mathrm{W\,m^{-1}K^{-1}}$, $L = 26.90$ cm).

Quantity	Insulated	Convecting	Difference
Θ_{tip}	0.5671	0.5607	insulated 1.1% high
$\dot{Q}_0/M$	0.8236	0.8280	insulated 0.53% low

TABLE III. Inputs for the provisional predictions of Parts 3 and 4.

Input	Value	Source
Rod diameter d	12.70 ± 0.02 mm	measured with digital calipers (0.01 mm resolution)
Overall rod length	28.66 ± 0.05 cm	measured with a ruler (1 mm divisions)
Hole diameter	2.50–2.55 mm (mean 2.525 mm)	measured with calipers
First-hole offset s	16.38 mm + $\frac{1}{2}$(2.525 mm) = 17.64 mm	calipers, from the TEC end of the rod to the near edge of the first hole, plus half the hole diameter to reach its center
Analyzed length L	28.66 cm − 1.764 cm = 26.90 ± 0.05 cm	first-hole center to the tip
Hole positions x	0, 1.2, 2.4, 3.7, 5.0, 7.6, 10.2, 15.3, 20.4 cm	measured from the first hole
Planned sensors	T1–T5 at $x = 0$, 5.0, 10.2, 15.3, 20.4 cm	chosen below
Position uncertainty δx	±1.3 mm	half the hole diameter: a sensor can sit anywhere in its hole
Side coefficient h	10 $\mathrm{W\,m^{-2}K^{-1}}$	Part III planning value for still room air [2]; not a measurement
Conductivity k	167 W/(m K); range 166–172 W/(m K)	6061-T6, Lienhard Table A.1: 20 °C value; range from the 0 °C and 100 °C values
Error criterion	$\epsilon_T < 5\%$ at every sensor used	Part III handout's suggested starting criterion

This gives $m = 4.343$ $\mathrm{m^{-1}}$, $1/m = 23.0$ cm, $mL = 1.168$, and $\mathrm{Bi}_\parallel = (mL)^2 = 1.364$. Over $k = 166$–172 W/(m K), $m = 4.279$–4.356 $\mathrm{m^{-1}}$ and $mL = 1.151$–1.172; this range changes the results below by much less than their rounding.

Error formula. Dividing the insulated-tip solution by the semi-infinite one,

$$\frac{\Theta_{\text{finite}}}{\Theta_\infty} = \frac{\cosh[m(L-x)]}{\cosh(mL)\, e^{-mx}} = \frac{1 + e^{-2m(L-x)}}{1 + e^{-2mL}} \;\geq\; 1,$$

so the insulated tip keeps the rod warmer than the semi-infinite model predicts, and the difference is largest near the tip. The relative error of the semi-infinite model is therefore

$$\epsilon_T(x) = \frac{\Theta_{\text{finite}} - \Theta_\infty}{\Theta_{\text{finite}}} = 1 - \frac{1 + e^{-2mL}}{1 + e^{-2m(L-x)}},$$

which increases monotonically from 0 at the base. Setting $\epsilon_T = 5\%$ gives the farthest position at which the criterion holds, $x^* = 5.4$ cm. Table IV gives ϵ_T at every hole and Table V at the planned sensors. Because h is only a planning value, Table VI varies it at $k = 167$ W/(m K). Values are from `python/analysis/a4_part3_finite_length.py`.

3.5 Prediction: length

With the criterion $\epsilon_T < 5\%$ at every sensor used, the semi-infinite model is adequate at T1 ($x = 0$) and T2 (5.0 cm, $\epsilon_T = 4.6\%$, still below 5% at the edge of the ±1.3 mm position uncertainty), but not at T3, T4, and T5 (10.2, 15.3, and 20.4 cm), where ϵ_T is 11%, 20%, and 30%. Among all the holes, only those at $x \leq 5.0$ cm meet the criterion ($x^* = 5.4$ cm). The whole rod is far from the "infinitely long" regime: $mL = 1.17$, compared with Lienhard's guideline $mL \gtrsim 5$.

Three of the five planned sensors lie beyond x^*, so I expect the course rod **not** to be adequately semi-infinite over the measured region. The stationary profile should be analyzed with the finite insulated-tip model; the semi-infinite fit is a comparison that should show a systematic residual at the far sensors. The tip boundary condition itself matters much less: replacing the insulated tip by a convecting one changes the prediction by about 1% (3.2).

This prediction depends on the planned sensor positions and $L = 26.90$ cm in 3.3, $h = 10$ $\mathrm{W\,m^{-2}K^{-1}}$, $k = 166$–172 W/(m K), and the 5% criterion. It is not sensitive to k. For any h from 5 to 20 $\mathrm{W\,m^{-2}K^{-1}}$, x^* stays between 4.6 and 7.4 cm, so T3, T4, and T5 fail in every case. Only T2 at 5.0 cm is marginal: it passes for $h \geq 10$ $\mathrm{W\,m^{-2}K^{-1}}$ but fails if h is as low as 5 $\mathrm{W\,m^{-2}K^{-1}}$.

PART 4. PROVISIONAL ONE-DIMENSIONAL PREDICTION

4.1 Transverse Biot number

Definition and meaning. The transverse (radial) Biot number is defined as (Lienhard Eq. 4.27)

$$\mathrm{Bi}_\perp = \frac{h(A/P)}{k},$$

where A/P, the cross-sectional area divided by the perimeter, is the characteristic length for conduction across the section. It represents the ratio of the internal radial conduction resistance to the external surface convection resistance:

$$\mathrm{Bi}_\perp \sim \frac{R_{\text{radial conduction}}}{R_{\text{surface convection}}}.$$

TABLE IV. Finite insulated-tip and semi-infinite profiles at every hole for $h = 10\ \mathrm{W\,m^{-2}K^{-1}}$, $k = 167\ \mathrm{W\,m^{-1}K^{-1}}$, $L = 26.90$ cm.

Hole x (cm)	$L - x$ (cm)	Θ_{finite}	Θ_∞	ϵ_T	ϵ_T for $x \pm \delta x$	$\epsilon_T < 5\%$?
0.0	26.90	1.0000	1.0000	0.00%	—	yes
1.2	25.70	0.9584	0.9492	0.96%	0.85–1.07%	yes
2.4	24.50	0.9194	0.9010	2.00%	1.89–2.12%	yes
3.7	23.20	0.8800	0.8516	3.23%	3.11–3.36%	yes
5.0	21.90	0.8434	0.8048	4.58%	4.44–4.71%	yes
7.6	19.30	0.7782	0.7189	7.62%	7.46–7.78%	no
10.2	16.70	0.7228	0.6421	11.16%	10.98–11.35%	no
15.3	11.60	0.6406	0.5146	19.67%	19.44–19.91%	no
20.4	6.50	0.5898	0.4123	30.09%	29.82–30.37%	no

TABLE V. Planned sensors, with an illustrative base excess temperature $\theta_0 = 25$ K.

Sensor	x (cm)	Θ_{finite}	$\ln \Theta_{\mathrm{finite}}$	θ (K)	ϵ_T
T1	0.0	1.0000	0.000	25.0	0.00%
T2	5.0	0.8434	−0.170	21.1	4.58%
T3	10.2	0.7228	−0.325	18.1	11.16%
T4	15.3	0.6406	−0.445	16.0	19.67%
T5	20.4	0.5898	−0.528	14.7	30.09%

TABLE VI. Sensitivity to the planning value of h at $k = 167\ \mathrm{W\,m^{-1}K^{-1}}$.

h ($\mathrm{W\,m^{-2}K^{-1}}$)	$1/m$ (cm)	mL	x^* (cm)
5	32.6	0.826	4.6
10	23.0	1.168	5.4
15	18.8	1.431	6.4
20	16.3	1.652	7.4

Form for a cylindrical rod. For a rod of diameter d,

$$\frac{A}{P} = \frac{\pi d^2/4}{\pi d} = \frac{d}{4}, \qquad \text{so} \qquad \boxed{\mathrm{Bi}_\perp = \frac{hd}{4k}}.$$

Numerical value and unit check. Using $h = 10\ \mathrm{W\,m^{-2}K^{-1}}$, $d = 0.01270$ m, and $k = 167\ \mathrm{W\,m^{-1}K^{-1}}$,

$$\mathrm{Bi}_\perp = \frac{(10)(0.01270)}{4(167)} = 1.90 \times 10^{-4}.$$

If k is taken from Lienhard's range of 166–172 W/(m K), then $\mathrm{Bi}_\perp = 1.85 \times 10^{-4}$ to 1.91×10^{-4}. Units:

$$\frac{[\mathrm{W\,m^{-2}K^{-1}}][\mathrm{m}]}{[\mathrm{W\,m^{-1}K^{-1}}]} = 1,$$

so $\mathrm{Bi}_\perp$ is dimensionless.

Conversion to a temperature difference. The temperature difference between the center of the cross section and its surface is approximately

$$\Delta T_{\mathrm{center-surface}} \approx \mathrm{Bi}_\perp\, \theta,$$

where θ is the temperature difference between the rod surface and the ambient air. With the illustrative $\theta \approx 25$ K of 3.4,

$$\Delta T \approx (2 \times 10^{-4})(25\ \mathrm{K}) \approx 0.005\ \mathrm{K},$$

about ten times smaller than the scatter of single thermistor readings (standard deviation 0.03–0.05 °C in the stationary interval of the rod runs).

Inverse calculation: what h would break the radial lumped assumption? Setting $\mathrm{Bi}_\perp = 0.1$,

$$h = \frac{0.4k}{d} = \frac{0.4(167)}{0.0127} \approx 5300\ \mathrm{W\,m^{-2}K^{-1}}.$$

Lienhard Table 1.1 (p. 21) lists 4.2–29 $\mathrm{W\,m^{-2}K^{-1}}$ for natural convection in air and 45–80 $\mathrm{W\,m^{-2}K^{-1}}$ for forced convection of air. The value 5300 $\mathrm{W\,m^{-2}K^{-1}}$ is about 500 times the planning value and still about 65 times the largest forced-air value. Under the experimental conditions, the assumption of radial temperature uniformity is therefore very robust.

4.2 Prediction: radius

Criterion. Lienhard (p. 24): the lumped-capacity solution is normally accurate within about 3% if $\mathrm{Bi} < 0.1$.

Conclusion. For this rod, $\mathrm{Bi}_\perp \approx 2 \times 10^{-4} \ll 0.1$. Therefore each cross section can be represented by a single temperature $T(x)$. This conclusion depends on $h = 10\ \mathrm{W\,m^{-2}K^{-1}}$, $k = 166$–172 W/(m K), and $d = 12.70$ mm.

Distinction from 3.4 and 3.5. $\mathrm{Bi}_\perp$ answers the approximation question in the **radial** direction; $\mathrm{Bi}_\parallel = (mL)^2$ in 3.4 and the prediction in 3.5 answer it in the **length** direction. They are independent: for this rod $\mathrm{Bi}_\perp$ is very small, but $(mL)^2 = 1.36$ is not. The temperature is uniform across each section, yet it still decays noticeably along the rod.

Status of this prediction. This is a prediction made *before* heating. After m is fitted, the self-consistency check

$$\mathrm{Bi}_\perp = \left(\frac{md}{4}\right)^2$$

will confirm whether the experimentally extracted m is consistent with a small $\mathrm{Bi}_\perp$.

4.3 Application: heat sink in still and forced air

Larger h in forced air. Forced airflow continuously sweeps away the heated air near the surface, thinning the thermal boundary layer, and thus significantly increases the convective heat transfer coefficient h. Lienhard Table 1.1 gives 4.2–29 $\mathrm{W\,m^{-2}K^{-1}}$ for natural convection in air and 45–80 $\mathrm{W\,m^{-2}K^{-1}}$ for forced convection of air (illustrative values; the actual h depends on velocity and geometry).

Relation between the decay length $1/m$ and h. From $m = \sqrt{hP/(kA)}$,

$$\frac{1}{m} = \sqrt{\frac{kA}{hP}} = \sqrt{\frac{kd}{4h}} \propto h^{-1/2}.$$

With $k = 167$ $\mathrm{W\,m^{-1}K^{-1}}$ and $d = 0.01270$ m, $1/m =$ 32.6, 23.0, 18.8, and 16.3 cm for $h = 5$, 10, 15, and 20 $\mathrm{W\,m^{-2}K^{-1}}$ (Table VI). When h is quadrupled, $1/m$ is halved. The forced-air heat sink therefore has the shorter decay length.

Physical meaning. A shorter decay length means that heat leaves through the sides before it can travel far along the fin, so the far end of the fin is closer to the ambient air temperature. The total heat dissipation,

$$\dot{Q}_0 = \sqrt{hPkA}\,\theta_0 \tanh(mL),$$

still increases with h. $\mathrm{Bi}_\perp \propto h$ also increases with h, but it remains $\ll 1$ in air.

GIT CHECKPOINT AND AI USE

This PDF is built from commit `94ede0416ea097463bf811f10fbb121f1848e46b` of `https://github.com/tyin-source/__Lab_Module_7__`; the same text is in `docs/assessments/a4_finite_length_biot.md`, and the LaTeX source in `docs/assessments/a4_latex/A4_Yin.tex`.

- **Part 1:** The derivation, the reasonableness discussion, and the choice of references are my own. Claude Code (an AI coding assistant) wrote `python/analysis/a4_problem_1_3.py`, which computes Table I from the Lienhard Table A.1 and A.2 values, reviewed my drafts, and formatted the references.
- **Part 2:** The calculations are my own. Claude Code organized and formatted this part: it wrote a first draft for comparison, reviewed my version, and fixed its LaTeX formatting.
- **Part 3:** The Problem 4.12 derivation, the Example 4.8 and Problem 4.20 calculations, and the choice of sensor holes with its reasons are my own. The course-rod numbers in 3.2–3.4 (m, mL, ϵ_T at each hole, x^*, the planned-sensor table, the tip comparison, and the h sensitivity) were computed by `python/analysis/a4_part3_finite_length.py`, which Claude Code wrote from my measurements; I checked m, mL, and ϵ_T at 7.6 cm by hand. Claude Code also organized and formatted Part 3: it made the tables, translated my notes into English, and drafted the text of 3.4 and 3.5.
- **Part 4:** The text and calculations are my own. Claude Code checked the numbers and the Lienhard citations against the textbook, corrected the page of the 3% statement (p. 24), the "two orders of magnitude" comparison, and the reading-noise value (now the measured scatter), and converted the math to GitHub format.
- **This PDF:** Claude Code converted the repository Markdown to this REVTeX document and drafted the abstract.

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[4] WebElements, "Silicon dioxide," `https://www.webelements.com/compounds/silicon/silicon_dioxide.html`.

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